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ORIGINAL PAPER

# Preliminary Trajectory Design for Mitigating Debris Utilizing Solar Sailing and Conventional Propulsion

**Kiduck Kim**[1]

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**Abstract**
In this study, preliminary trajectory design for debris removal in low Earth orbit using solar sails is explored. Emphasis is placed on regions below 1000 km altitude, where the debris population is rapidly growing. The aim is to propose a mission concept capable of repetitively executing removal processes. The trajectory is intricately crafted, segmented into rendezvous, proximity operations, and deorbiting phases. Safety is prioritized by leveraging walking safety ellipses during proximity operations, ensuring efficient capture of targeted debris. Additionally, the feasibility and limitations of the mission concept are assessed through numerical simulations based on characteristic acceleration, a pivotal performance index of solar sails.

**Keywords** Debris removal · Deorbit · Proximity operation · Rendezvous · Solar sail

## 1 Introduction

The concept of solar sails was first introduced in the early tentieth century [1]. However, it is only in recent years that practical demonstrations in space have become feasible, largely due to the intricate engineering challenges associated with the storage, deployment, and management of thin, large-area sails, along with their accompanying mechanisms and mechanical devices [2–4]. A standout feature of solar sailing is its ability to generate propulsion without the need for fuel, a characteristic that has attracted consistent interest from both the scientific community and NASA for potential future missions. This propellant-free approach offers several advantages, including the reduction of spacecraft weight by eliminating the need for fuel, thereby reducing launch costs, while also enabling increased payload capacity. Furthermore, opportunities for missions involving high-energy concepts that are previously difficult to execute due to the substantial del-$V$ requirements of conventional propulsion methods are opened up by this approach. Such missions include rendezvous with near-Earth asteroids [5], sample return missions from terrestrial planets [6–8], solar polar imager missions for monitoring space weather changes [9], and interplanetary missions [10].

Discussions regarding the application and scope of solar sails in mission execution continue to be thoroughly discussed and explored. Recently, proposals have emerged suggesting the utilization of solar sails as a propulsion system for debris removal or mitigation [11, 12]. While traditional debris removal typically involves a single spacecraft targeting one piece of debris, strategies proposing the removal of multiple debris have been put forward to significantly reduce mission costs [13, 14]. Within this context, solar sails, unlike conventional chemical propulsion methods requiring propellant, are being seriously considered as a feasible alternative. Additionally, the majority of debris in low Earth orbit is concentrated between 600 and 1000 km altitude [15, 16], with debris below 1000 km altitude showing a rapid increase [17]. Consequently, this study aims to design trajectories for debris removal in low Earth orbit using solar sails, with a specific focus on the perspective of multiple debris removal.

Trajectory design based on solar sails can be categorized into two main approaches: sun centered [18] and planet centered [19]. Sun-centered trajectories are commonly applied to interplanetary missions where Earth's gravity is often not a significant factor, while planet-centered trajectories are used to address issues within Earth's orbit. Planet-centered trajectories, due to dealing with a relatively more complex perturbed environment in dynamics, were initially limited to relatively simple missions such as orbit inclination changes

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✉ Kiduck Kim
KiduckKim@kari.re.kr

[1] Korea Aerospace Research Institute, Daejeon, Korea

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[20], Earth escape [21], and lunar flybys [22]. Moreover, consideration of atmospheric drag is crucial when designing trajectories within the Earth's orbit. While the atmosphere is nearly negligible above altitudes of 1000 km, atmospheric drag must be addressed as a perturbing force when designing trajectories below this altitude [23]. However, despite this, there have been instances where the influence of the atmosphere was entirely disregarded during the problem definition process, or if considered, the representation of the sail's configuration was simplified and not fully integrated [24–26].

In this regard, this study aims to develop trajectories for mitigating debris in low Earth orbit, while considering the influence of atmospheric drag. Existing literature on solar sail-based approaches has primarily focused on optimizing rendezvous trajectories, either through the application of novel optimization techniques or by redefining analytical models for this purpose [27]. However, the practical implementation of active debris removal strategies, such as capture, necessitates additional considerations beyond those addressed in existing literature. In this context, proximity operations involve trajectories that allow mission execution on the target satellite or debris by gradually reducing relative velocity while approaching. Therefore, the primary contribution of this paper lies in the proposal of trajectories for both rendezvous and proximity operations for multiple debris removal, along with an analysis of the feasibility and limitations of such challenging missions based on key mission-related parameters.

This manuscript examines the concept of multiple debris removal by implementing a sequence of orbit transfers, proximity operations, and deorbit missions for a single target, and then repeating this process for additional targets. As such, the study emphasizes the design of mission trajectories for individual missions while also analyzing the necessary conditions for repeating these processes through numerical simulations. To accomplish this, the trajectory design is divided into two primary phases: orbit transfer and proximity operation. In the orbit transfer phase, which involves significant fuel consumption, the use of solar sails is explored exclusively. For the proximity operation phase, we have introduced a walking safety ellipse to enhance passive safety and opted for a conventional thruster for impulsive burns, due to its superior thrust capability compared to the relatively modest thrust of the solar sail. As a result, in the orbit transfer phase, the analysis of orbit raising and lowering capabilities helps to determine the characteristic acceleration values required for the solar sail, which are essential for the repeated execution of a single mission. This characteristic acceleration, a crucial performance metric for solar sails, allows for the estimation of the sail area and spacecraft mass needed for effective multiple debris removal missions. Furthermore, in the proximity operation phase, a summary of the delta-$V$ requirements for each stage, along with the total delta-$V$ necessary for a single mission, is provided to assist in the selection of appropriate thrusters.

The paper is structured as follows. In Sect. 2, a concise background is provided for readers unfamiliar with solar sail-based spacecraft, along with proposed mission concepts for multiple debris removal. Section 3 explains the mathematical models utilized in trajectory design, covering not only spacecraft dynamics but also the accelerations induced by solar sails, Earth's gravitational imbalance, and atmospheric drag disturbances. Section 4 conducts numerical simulations. Parametric studies are performed by varying the characteristic acceleration, considered a fundamental performance metric of solar sails, and the resulting outcomes are thoroughly discussed. Finally, in Sect. 5, key conclusions are drawn and suggestions for improvements and future research directions are outlined.

## 2 Background

### 2.1 Operation Concept

Here, an explanation of the mission concept is provided for readers unfamiliar with debris removal tasks. This study examines trajectory design for multiple debris removal missions based on solar sail technology, with Fig. 1 illustrating the relevant mission concept. Initial operations are performed by the spacecraft and the solar sail is deployed in a parking orbit, maintaining altitude while awaiting the selection of target debris from the ground. If a target is selected, a transfer orbit is required for capture. In contrast to existing literature, which mainly focuses on presenting orbit rendezvous, proximity operations are additionally considered alongside orbit rendezvous in our study. To execute practical actions such as capture, proximity operations trajectories need to be generated to ensure that the spacecraft can remain near the target for inspection and perform necessary maneuvers. This is because with only orbit rendezvous, the relative distance and velocity between the spacecraft and target are only suitable for brief flyby-type trajectories, allowing for capture and similar operations for a very short duration.

Following proximity operations, once the target has been successfully captured, the spacecraft is directed to lower its altitude with the target to a disposal orbit. The purpose of this disposal orbit is to support the 25-year rule proposed by the Inter-Agency Space Debris Coordination Committee (IADC) for the mitigation of orbital debris. According to this rule, artificial objects in low Earth orbit should be removed from orbit within 25 years to prevent them from posing a debris hazard to other spacecrafts [28]. The disposal orbit is assumed to be at an altitude of 550 km. While the exact duration may vary slightly depending on the size and weight of the spacecraft, without additional devices related to deorbiting,

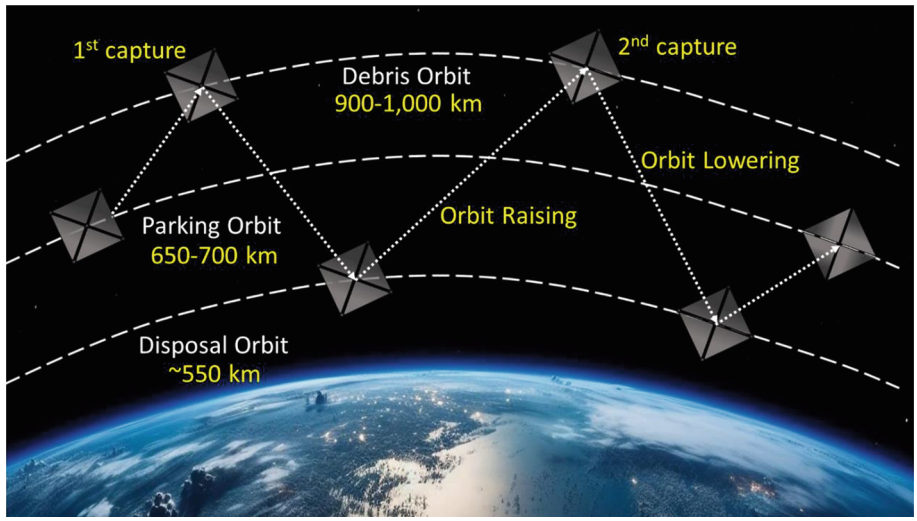


**Fig. 1** Concept of multiple debris removal operations utilizing solar sails

the natural decay time typically satisfies the 25-year limit at an altitude of around 600 km [29]. Therefore, summarizing the mission, a series of processes is repetitively performed by the spacecraft, involving the capture of debris at altitudes higher than 600 km and their transportation to a disposal orbit.

The difference between the proposed mission concept in debris removal missions and conventional approaches lies in the more active utilization of solar sails. Traditional utilization of solar sails in debris removal missions has been passive, with them being used in the form of drag devices. Consequently, in previous missions, debris is approached using conventional propulsion systems for capture, followed by deploying the sail to increase drag and facilitate orbit disposal through enlarged cross-sectional area.

However, solar sails offer not only orbit lowering but also the capability of raising it. By substituting the relatively fuel-intensive altitude ascent with solar sails, fuel consumption for a single mission can be reduced, enabling repetitive missions. It is crucial to note that in this study, the designed trajectory doesn't solely rely on solar sails as the exclusive propulsion system. Although solar sails are employed throughout orbit-raising and -lowering processes, conventional propulsion systems are used to create proximity operation trajectories. This is because the thrust generated by solar sails is inherently small, making it challenging to generate the required del-$V$ for creating proximity operation trajectories. A more detailed analysis of the designed trajectory will be provided in Sect. 4, along with numerical values.

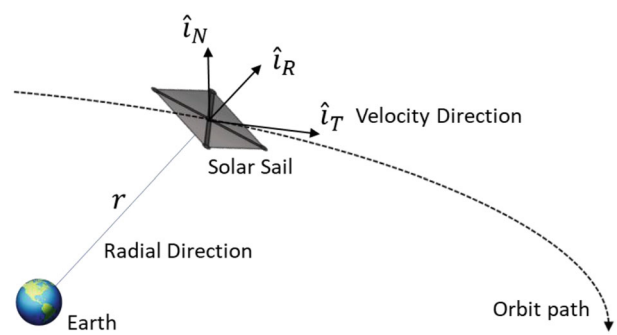


**Fig. 2** Definition of the radial–transverse–normal frame

## 2.2 Solar Radiation Pressure

The propulsive acceleration of the solar sail is generated by solar radiation pressure and can be defined as follows.

$$\boldsymbol{a}_{\mathrm{SRP}} = a_c \left( \frac{r_\odot}{\boldsymbol{r}_s} \right)^2 \left( \boldsymbol{X}_s \cdot \hat{\boldsymbol{n}}_s \right)^2 \hat{\boldsymbol{n}}_s. \tag{1}$$

Here, $\boldsymbol{a}_{\mathrm{SRP}} \in \mathbb{R}^{3\times 1}$ represents the vector expressed in the typical radial–transverse–normal (RTN) frame of unit vectors $\left\{ \hat{\imath}_{\mathrm{R}}, \hat{\imath}_{\mathrm{T}}, \hat{\imath}_{\mathrm{N}} \right\}$, indicating the acceleration induced by the solar sail. In the RTN frame, $\hat{\imath}_{\mathrm{N}}$ is typically denoted as the direction of orbital angular momentum, $\hat{\imath}_{\mathrm{R}}$ represents the direction from the center of the Earth to the spacecraft, and $\hat{\imath}_{\mathrm{T}}$ is determined by the right-handed rule of orthogonal coordinates, representing the direction of velocity in orbit [30]. These definitions are illustrated in Fig. 2.

The term $a_{\mathrm{c}}$ is indicative of the characteristic acceleration, representing the magnitude of the propulsive acceleration generated when the spacecraft is positioned at a distance of 1 astronomical unit (au) from the Sun, with incident sunlight striking the solar sail perpendicularly. Here, $r_\odot$ signifies the

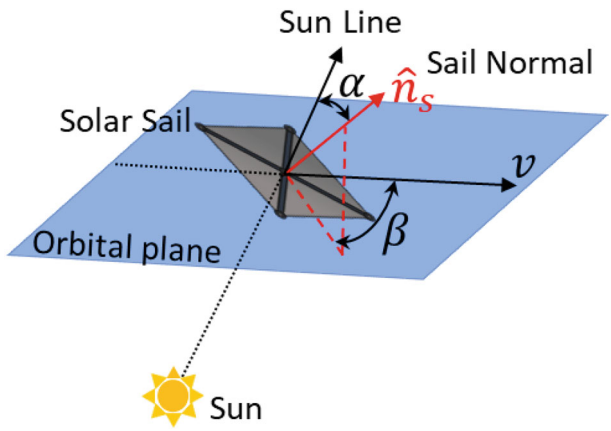


**Fig. 3** Illustration of the cone angle and clock angle of the solar sail

reference distance of 1 au from the Sun, while $\boldsymbol{r}_s$ denotes the distance between the spacecraft and the Sun. Therefore, the magnitude of $\boldsymbol{a}_{\mathrm{SRP}}$ is directly proportional to $a_{\mathrm{c}}$ and inversely proportional to the square of the distance from the Sun. $\boldsymbol{X}_s$ and $\hat{\boldsymbol{n}}_s$, respectively, represent the direction from the Sun to the spacecraft and the normal vector of the solar sail. The orientation of the solar sail's normal vector can be defined by two angles: the cone angle $\alpha$ and the clock angle $\beta$. Definitions for these two angles to define the normal vector can be confirmed from Fig. 3.

Therefore, the normal vector $\hat{\boldsymbol{n}}_s$ can be defined as follows:

$$\hat{\boldsymbol{n}}_s = \left[\cos\alpha,\ \sin\alpha\cos\beta,\ \sin\alpha\sin\beta\right]^T. \tag{2}$$

Additionally, since the dot product of $\boldsymbol{X}_s \cdot \hat{\boldsymbol{n}}_s$ equals the cosine value of the angle $\alpha$ between them, $\boldsymbol{a}_{\mathrm{SRP}}$ can be expressed as follows:

$$\boldsymbol{a}_{\mathrm{SRP}} = a_{\mathrm{c}}\left(\frac{r_{\odot}}{\|\boldsymbol{r}_s\|}\right)^2 \cos^2\alpha \begin{pmatrix} \cos\alpha \\ \sin\alpha\cos\beta \\ \sin\alpha\sin\beta \end{pmatrix}. \tag{3}$$

The size of $\boldsymbol{a}_{\mathrm{SRP}}$, a crucial performance parameter of a solar sail, is determined by the magnitude of $a_{\mathrm{c}}$. A higher $a_{\mathrm{c}}$ indicates a stronger acceleration generated by the solar sail, facilitating more efficient orbit adjustments. This value is directly proportional to the sail's area, but inversely proportional to the spacecraft's mass. Specifications for solar sails designed for actual flight missions, such as IKAROS [31], NanoSail-D2 [32], LightSail 1 [33], LightSail 2 [4, 34], and NEA Scout [35], are detailed in Table 1. It is noteworthy that the ratio of sail area to mass progressively increases, aiming to optimize the small thrust of solar sails, and consequently leading to an escalation in the $a_{\mathrm{c}}$ value. Since the inception of LightSail 1 in 2015, spacecraft have maintained $a_{\mathrm{c}}$ values around 0.06 mm/s$^2$, with the potential for further increase with larger sail areas in the future.

## 3 Dynamics

### 3.1 Modified Equinoctial Orbital Elements (MEOEs)

The dynamics of spacecraft movement in orbit are characterized by the use of modified equinoctial elements (MEOEs). Introduced in [37, 38], MEOEs serve to represent various types of osculating orbits as a set of orbital elements. These values can be derived through straightforward calculations from Keplerian elements, known as classical orbital elements (COEs), and are recognized for their suitability in perturbation analysis and trajectory optimization. If classical orbital elements are designated as $\left\{a,\ e,\ i,\ \omega,\ \Omega\ \nu\right\}$, representing semi-major axis, eccentricity, inclination, argument of periapsis, longitude of the ascending node, and true anomaly, respectively, then MEOEs $\left\{p,\ f,\ g,\ h,\ \mathrm{k}\ L\right\}$ can be defined accordingly.

$$p = a\left(1 - e^2\right), \tag{4}$$

$$f = e\cos(\omega + \Omega), \tag{5}$$

$$g = e\sin(\omega + \Omega), \tag{6}$$

$$h = \tan\left(\frac{i}{2}\right)\cos\Omega, \tag{7}$$

$$k = \tan\left(\frac{i}{2}\right)\sin\Omega, \tag{8}$$

**Table 1** Specifications of solar sails in real flight missions

| | IKAROS | NanoSail-D2 | LightSail 1 | LightSail 2 | NEA Scout |
|---|---|---|---|---|---|
| Developed by | JAXA | NASA MSFC | The planetary Society | The planetary Society | NASA MSFC/JPL |
| Launch year | 2010 | 2010 | 2015 | 2019 | 2022 |
| Area (m$^2$) | 196 | 10 | 32 | 32 | 86 |
| Mass (kg) | 322 | 5.35 | 5.38 | 5.58 | 14 |
| $a_c$(mm/s$^2$) | 5.8e−3 | 2.3e−2 | 5.9e−2 | 5.8e−2 | 6.3e−2 |

$$L = \nu + \Omega + \omega. \tag{9}$$

If the state vector $\boldsymbol{x}$ is defined by MEOEs, the differential equation that defines the spacecraft's dynamics can be organized in the following matrix form [39].

$$\boldsymbol{x} = \left\{ p,\ f,\ g,\ h,\ \mathrm{k}\ L \right\}^{\boldsymbol{T}}, \tag{10}$$

$$\dot{\boldsymbol{x}} = \mathbf{A}(\boldsymbol{x})\boldsymbol{a}_{\mathrm{RTN}} + \boldsymbol{B}(\boldsymbol{x}). \tag{11}$$

Here, $\boldsymbol{a}_{\mathrm{RTN}} \in \mathbb{R}^{3\times 1}$ denotes the propulsive vector expressed in the RTN frame. $\boldsymbol{a}_{\mathrm{RTN}}$ encompasses perturbation terms, including additional forces due to solar sail, gravitational imbalances, drag, and lift induced by the atmosphere. The matrices $\mathbf{A}(\boldsymbol{x}) \in \mathbb{R}^{6\times 3}$ and $\boldsymbol{B}(\boldsymbol{x}) \in \mathbb{R}^{6\times 1}$ are provided as follows, with $\mu_{\odot}$ representing the gravitational constant:

$$\mathbf{A}(\boldsymbol{x}) = \sqrt{\frac{p}{\mu_{\odot}}} \begin{bmatrix} 0 & \frac{2p}{q} & 0 \\ \sin L & \frac{(1+q)\cos L + f}{q} & -\frac{g(h\sin L - k\cos L)}{q} \\ -\cos L & \frac{(1+q)\sin L + f}{q} & \frac{f(h\sin L - k\cos L)}{q} \\ 0 & 0 & \frac{(1+h^2+k^2)\cos L}{2q} \\ 0 & 0 & \frac{(1+h^2+k^2)\sin L}{2q} \\ 0 & 0 & \frac{h\sin L - k\cos L}{q} \end{bmatrix}, \tag{12}$$

$$\boldsymbol{B}(\boldsymbol{x}) = \sqrt{\mu_{\odot} p} \begin{bmatrix} 0_{5\times 1} \\ \left(\frac{q}{p}\right)^2 \end{bmatrix}, \tag{13}$$

where

$$q = 1 + f\ \cos L + g \sin L. \tag{14}$$

Celestial bodies, including Earth, are not perfect spheres, but rather assume a slightly elongated shape with a larger radius at the equator than at the poles due to the centrifugal force. Consequently, an uneven distribution of gravity occurs, resulting in varying gravitational forces acting on spacecraft depending on their position. This uneven gravitational field distribution must be considered in the equations of motion, especially in low Earth orbit. The perturbation experienced by spacecraft due to the uneven gravitational field of Earth is typically modeled using the zonal harmonics coefficient, $J_2$, and is sometimes referred to as the $J_2$ effect or $J_2$ perturbation. Here, $J_2$ represents a dimensionless parameter and can be expressed using modified equinoctial elements in the RTN frame as follows:

$$\boldsymbol{a}_{J_2} = \begin{pmatrix} -\frac{3\mu_{\odot} J_2 R_{\odot}^2}{2\|r\|^4}\left[1 - \frac{12(h\sin L - k\cos L)^2}{(1+h^2+k^2)^2}\right] \\ -\frac{12\mu_{\odot} J_2 R_{\odot}^2}{\|r\|^4}\left[\frac{(h\sin L - k\cos L)(h\cos L + k\sin L)}{(1+h^2+k^2)^2}\right] \\ -\frac{6\mu_{\odot} J_2 R_{\odot}^2}{\|r\|^4}\left[\frac{(1-h^2-k^2)(h\sin L - k\cos L)}{(1+h^2+k^2)^2}\right] \end{pmatrix}. \tag{15}$$

Here, $\|r\|$ and $R_{\odot}$, respectively, represent the distance between the Earth's center and the spacecraft, and the mean equatorial radius of the Earth.

Additionally, due to the large cross-sectional area of the sail, aerodynamic perturbations from drag and lift caused by the atmosphere must be considered in low Earth orbit. Assuming the sail to be a flat plate, the aerodynamic acceleration can be modeled as follows [23]:

$$\boldsymbol{a}_{\mathrm{Aero}} = \frac{1}{2}\frac{A_s}{m_s}\rho v^2\left(C_L \hat{\boldsymbol{v}}_{\perp} - C_D \hat{\boldsymbol{v}}\right). \tag{16}$$

Here, $A_s$, $m_s$, and $\rho$ are, respectively, the sail area, spacecraft mass, and local atmospheric density, while $v$ and $\hat{\boldsymbol{v}}$ denote the magnitude and direction vector of the spacecraft's velocity. $\hat{\boldsymbol{v}}_{\perp}$ denotes the lift direction vector and is defined as follows [40]:

$$\hat{\boldsymbol{v}}_{\perp} = \mathrm{sign}\left(\hat{\boldsymbol{v}} \cdot \hat{\boldsymbol{n}}_s\right)\frac{\hat{\boldsymbol{v}} \times \left(\hat{\boldsymbol{v}} \times \hat{\boldsymbol{n}}_s\right)}{\hat{\boldsymbol{v}} \times \hat{\boldsymbol{n}}_s}. \tag{17}$$

Here, $\mathrm{sign}(a)$ is a function that returns the sign of $a$. The lift and drag coefficients, denoted by $C_{\mathrm{L}}$ and $C_{\mathrm{D}}$, can be defined as follows:

$$C_{\mathrm{L}} = 2[(2 - \sigma_{\mathrm{N}} - \sigma_{\mathrm{T}})|\cos\xi| + \sigma_{\mathrm{N}} V_{\mathrm{R}}]\,|\cos\xi|\ \sin\xi, \tag{18}$$

$$C_{\mathrm{D}} = 2\left[\sigma_{\mathrm{T}} + \sigma_{\mathrm{N}} V_{\mathrm{R}}|\cos\xi| + (2 - \sigma_{\mathrm{N}} - \sigma_{\mathrm{T}})\cos^2\xi\right]|\cos\xi|. \tag{19}$$

$\xi$ represents the angle between the normal vector of the sail plane and the velocity direction, indicating the angle of attack of the sail in the direction of its velocity, as described in Fig. 4. $V_{\mathrm{R}}$ denotes the ratio between the thermal speed of atmospheric particles and the spacecraft velocity, while $\sigma_{\mathrm{T}}$ and $\sigma_{\mathrm{N}}$, respectively, represent the tangential and normal momentum accommodation coefficients. Utilizing the values from [23], $V_{\mathrm{R}}$ is set to 0.05, and $\sigma_{\mathrm{T}}$ and $\sigma_{\mathrm{N}}$ are both set to 0.8. Consequently, $\boldsymbol{a}_{\mathrm{RTN}}$ can be defined as follows:

$$\boldsymbol{a}_{\mathrm{RTN}} = \boldsymbol{a}_{\mathrm{SRP}} + \boldsymbol{a}_{J_2} + \boldsymbol{a}_{\mathrm{Aero}}. \tag{20}$$

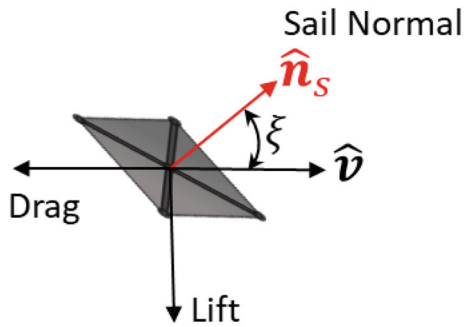


**Fig. 4** Illustration of aerodynamic perturbation in the solar sail

### 3.2 Control Strategy

The control parameters for designing the trajectory of a solar sail are determined by the cone angle $\alpha$ and the clock angle $\beta$. These angles cause alterations in the normal vector of the sail plane, thereby inducing orbit variations. In cases such as low Earth orbit where various environmental perturbation terms must be considered, locally optimal control laws have been utilized to avoid complexity and ensure convergence [41]. This method aims to maximize changes in specific orbit elements at each time step, inducing the desired trajectory variations, and has been widely employed in the literature [18, 19, 23, 41]. Therefore, in this manuscript, the emphasis was placed on designing trajectories using the concept of locally optimal control, rather than comparing the optimality of trajectories designed through global optimal control laws and assessing the feasibility of the proposed multiple debris removal operational concept.

As noted in Sect. 2, in dynamics where various perturbation terms are considered, such as in low Earth orbit, trajectory design is executed through the concept of locally optimal solutions. Trajectory design essentially entails adjusting the spacecraft's current orbital elements to target orbital elements, and the concept of locally optimal solutions facilitates orbit transfers by maximizing changes in specific orbital elements at each time step. Here, specific orbital elements can be individual elements or a blend of multiple elements, enabling simultaneous control over several orbital elements [18, 19]. The controlled orbital elements primarily encompass semi-major axis, eccentricity, and inclination. The rates of change for these elements over time can be expressed through the relationship between MEOEs and COEs:

$$\frac{\mathrm{d}a}{\mathrm{d}t} = \left(\frac{1}{1-f^2-g^2}\right)\frac{\mathrm{d}p}{\mathrm{d}t} + \left(\frac{2p}{\left(1-f^2-g^2\right)^2}\right)\left(f\frac{\mathrm{d}f}{\mathrm{d}t} + g\frac{\mathrm{d}g}{\mathrm{d}t}\right), \tag{21}$$

$$\frac{\mathrm{d}e}{\mathrm{d}t} = \left(\frac{f}{\sqrt{f^2+g^2}}\right)\frac{\mathrm{d}f}{\mathrm{d}t} + \left(\frac{g}{\sqrt{f^2+g^2}}\right)\frac{\mathrm{d}g}{\mathrm{d}t}, \tag{22}$$

$$\frac{\mathrm{d}i}{\mathrm{d}t} = \left(\frac{2}{\left(1+h^2+k^2\right)\sqrt{h^2+k^2}}\right)\left(h\frac{\mathrm{d}h}{\mathrm{d}t} + k\frac{\mathrm{d}k}{\mathrm{d}t}\right). \tag{23}$$

In this case, the score function for achieving local optimality can be formulated in a blended law format.

$$S_{\mathrm{tot}} = W_a S_a + W_e S_e + W_i S_i, \tag{24}$$

$$S_a = \left(\frac{\left(a_{\mathrm{target}} - a\right)}{a_0^2}\right)\frac{\mathrm{d}a}{\mathrm{d}t}, \tag{25}$$

$$S_e = \left(e_{\mathrm{target}} - e\right)\frac{\mathrm{d}e}{\mathrm{d}t}, \tag{26}$$

$$S_i = \left(i_{\mathrm{target}} - i\right)\frac{\mathrm{d}i}{\mathrm{d}t}. \tag{27}$$

Here, $a_{\mathrm{target}}$, $e_{\mathrm{target}}$, and $i_{\mathrm{target}}$, respectively, denote the target values for each orbital element to be reached through control inputs. $a_0$ represents the initial value of each semi-major axis, serving as a scaling factor for differences in magnitude with other orbital elements. Control performance metrics, such as the raising time to approach the set target values for each orbital element, can vary depending on the weighting factors $W_a$, $W_e$, and $W_i$ applied to the score function. The primary goal of the local optimal controller for rendezvous missions is to ensure that the three orbital elements achieve their target values accurately within the designated mission or transfer time. However, when adjusting orbital elements using a solar sail, the rates of change for these elements, as described by the blended law, vary. Specifically, changes in the semi-major axis typically occur more rapidly compared to changes in eccentricity and inclination. If the same weight factors are applied across all elements, the results may be dominated by the semi-major axis, potentially leading to slower convergence of eccentricity and inclination to their target values. To address this, the weight factor for the semi-major axis, $W_a$, is set smaller than those for eccentricity and inclination, $W_e$, and $W_i$, to more accurately reflect the relative difficulty in modifying eccentricity and inclination.

In this study, an analysis of the performance changes with variations in these values was not conducted, and fixed values of $\left[W_a,\ W_e,\ W_i\right] = \left[0.1,\ 1,\ 1\right]$ were used throughout the entire simulation time. Consequently, at each time step, $S_{\mathrm{tot}}$ is maximized to maximize changes in the orbital elements comprising it, determining the cone angle $\alpha^*$ and clock angle $\beta^*$.

$$\arg\max_{\alpha,\ \beta}\{S_{\mathrm{tot}}\}.$$

**Table 2** Initial conditions for spacecraft and target debris

| | Semi-major axis (km) | Eccentricity | Inclination (deg) |
|---|---|---|---|
| Spacecraft initial | 6978.14 | 0.01 | 95 |
| Target debris | 7378.14 | 0.0001 | 95.5 |

When the control parameters of the solar sail are determined by $\alpha^*$ and $\beta^*$, the corresponding normal vector can be computed, allowing for the calculation of $\boldsymbol{a}_{\mathrm{SRP}}$. This value, along with $\boldsymbol{a}_{J_2}$ and $\boldsymbol{a}_{\mathrm{Aero}}$ calculated at that time instant, is used to compute $\boldsymbol{a}_{\mathrm{RTN}}$ and is incorporated into the orbit propagation through the dynamics of MEOEs.

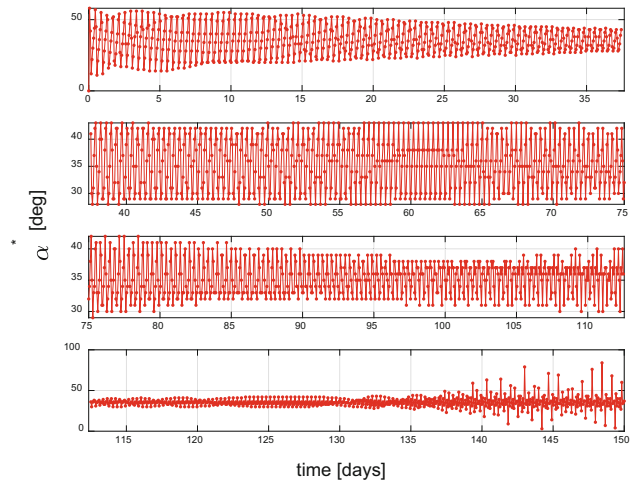


**Fig. 6** Changes in the optimized cone angle $\alpha^*$ over time

## 4 Numerical Simulations

### 4.1 Rendezvous Phase

The rendezvous phase involves transferring the orbit of a spacecraft, situated in a parking orbit as described earlier, to the debris orbit. The semi-major axis of the parking orbit is assumed to be 6978.14 km, with eccentricity and inclination values of 0.01 and 95 degrees, respectively. The target debris orbit has a semi-major axis of 7378.14 km, with assumed eccentricity and inclination values of 0.0001 and 95.5 degrees, respectively, as summarized in Table 2.

The spacecraft's specifications are benchmarked against those of the recently developed NEA Scout, with assumed sail area and spacecraft mass values of 86 $m^2$ and 14 kg, respectively, resulting in a characteristic acceleration of 0.063 mm/$s^2$. The time step for numerical integration is set to 100 s, and the optimized angle through the score function is recalculated and reflected every 100 s as well. The results of the rendezvous operation are depicted in Fig. 5. It is observed that the spacecraft's semi-major axis, eccentricity, and inclination, utilizing only the solar sail, approach the target values after approximately 146 days and maintain these values upon reaching the target.

Figures 6 and 7 depict the optimized angles α* and β* calculated for controlling the sail. In this study, preliminary trajectory design for mission planning is focused on, with no consideration given to the attitude control loop for spacecraft equipped with a sail. It is currently assumed that the calculated optimized angles are accurately followed, with the spacecraft requiring maximum rates of approximately

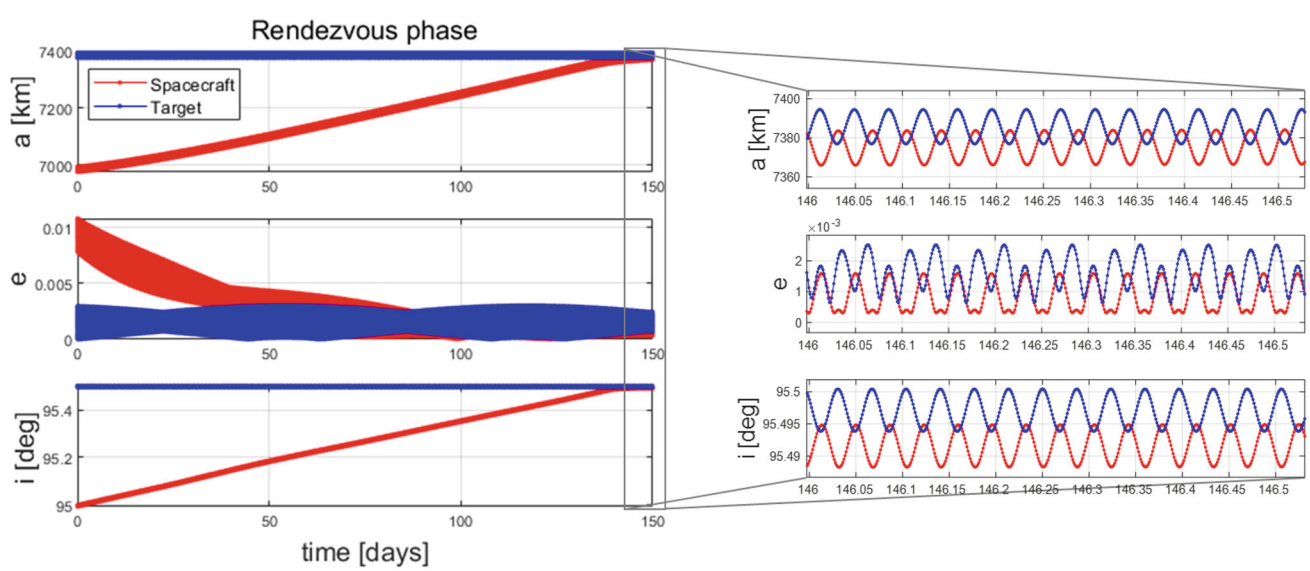


**Fig. 5** Results of rendezvous phase simulation using local optimal control

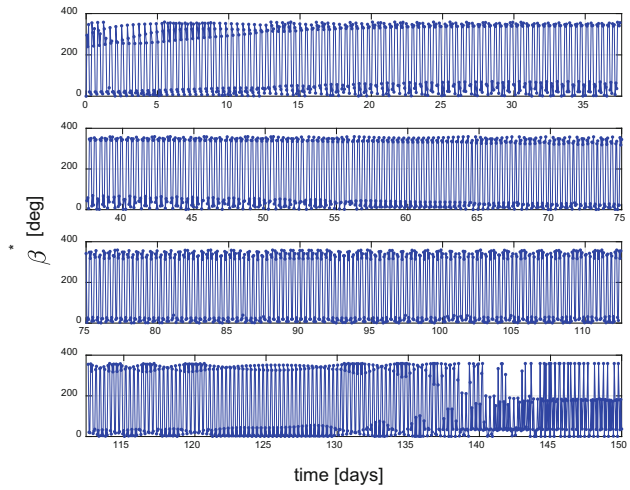


**Fig. 7** Changes in the optimized clock angle $\beta^*$ over time

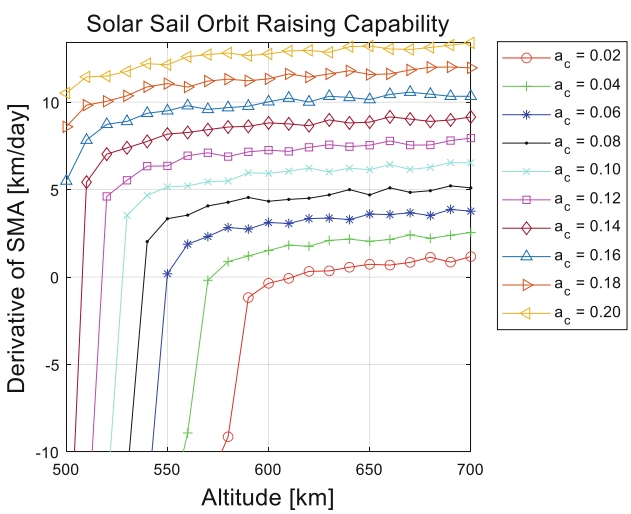


**Fig. 8** Availability of orbit raising for solar sail based on changes in $a_c$

18 deg/min for α* and 21 deg/min for β* for sail orienting. For more realistic designs, considerations of control capabilities, delays, and accuracy need to be incorporated.

Solar sail-based orbit raising varies depending on the size of the performance parameter $a_c$. Here, performance does not solely refer to the orbit-raising speed required to approach the target. Due to the minimal magnitude of $\boldsymbol{a}_{SRP}$ generated by the sail, there is a critical altitude beyond which the wide-area sail cannot overcome the drag as the spacecraft's altitude decreases, rendering altitude ascent impossible. Therefore, an analysis of the threshold altitude based on the size of $a_c$ is necessary. For this purpose, the values of $W_e$ and $W_i$ are set to 0 in the score function to focus solely on altitude ascent, and the average altitude changes are compared while varying $a_c$. Figure 8 illustrates the results.

The threshold altitude, where altitude ascent becomes impossible, is identified as the point where the altitude change

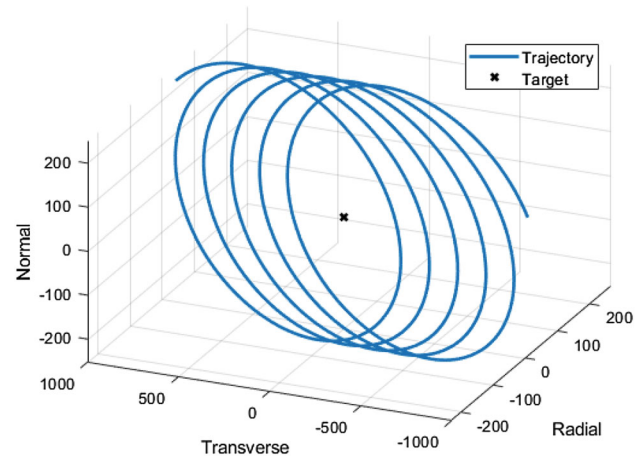


**Fig. 9** Example of relative trajectory for walking safety ellipse

is zero. It is observed that the threshold altitude is 550 km when $a_c$ is 0.06 mm/s$^2$. Consequently, when altitude ascent using only the sail is considered in mission design, the starting altitude must be set higher than the threshold altitude. Moreover, the results indicate that the threshold altitude tends to increase as the size of $a_c$ increases. While increasing the sail area is an easy way to enlarge $a_c$, difficulties in storage and deployment arise with larger sail areas. Therefore, appropriately selecting the required size of $a_c$ based on mission objectives beforehand is crucial.

## 4.2 Proximity Operation Phase

In this study, unlike other solar sail-related research, we consider the generation of post-rendezvous proximity operation trajectories, designed with passive safety in mind, using the walking safety ellipse (WSE). Here, passive safety implies that the trajectory allows the spacecraft to pass by the target without collision if no specific action is taken for final approaches. Among various options, the WSE best fits this requirement. As depicted in Fig. 9, the WSE forms a spiral-shaped manifold in the velocity direction, or the transverse direction, in the RTN frame. It manifests as an ellipse in the radial–normal plane with the target at the center. At the point of closest approach to the target, the spacecraft can pass by without the risk of collision, even without any maneuvers. More detailed explanations and mathematical support for the WSE can be found in existing literature [42–44].

After the orbit transfer near the target using the sail, maneuvers for proximity operations are carried out step by step, with the entire trajectory depicted in Fig. 10. Burn #1 denotes the maneuver to shift to the relative position for WSE injection post-orbit transfer, depicted as WSE transfer. Burn #2 is a maneuver to align with the required relative velocity condition for WSE injection at the injection point. To

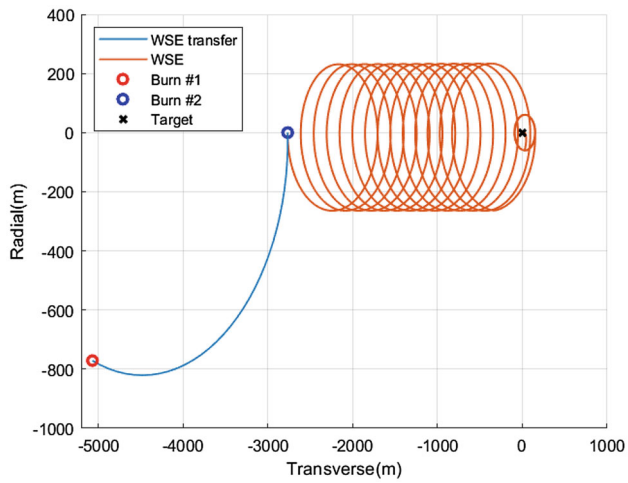


**Fig. 10** The WSE trajectory generated after orbit transfer using the sail

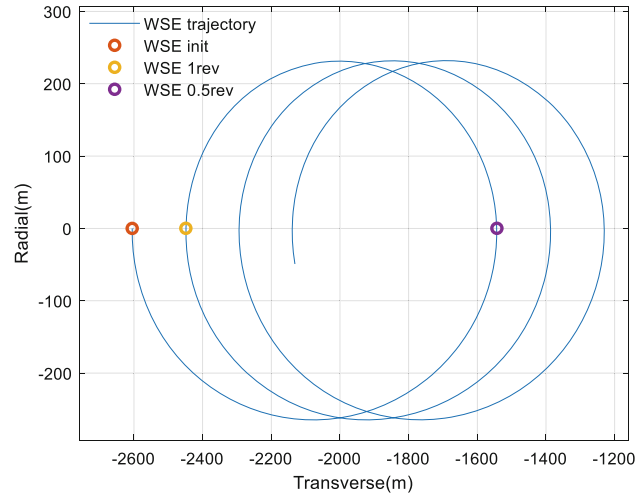


**Fig. 11** Description of walking rate

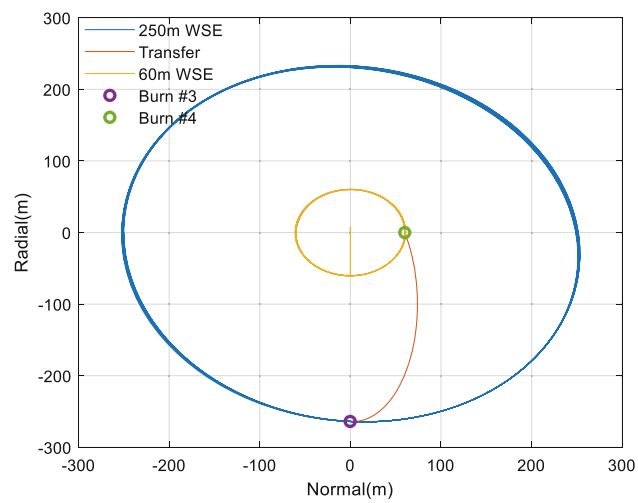


**Fig. 12** Transfer and injection for generating reduced-sized WSE

perform WSE injection, the relative velocity with the target needs to be adjusted to a specific value. However, due to the small magnitude of the $\boldsymbol{a}_{\mathrm{SRP}}$, providing the required del-$V$ for WSE injection is challenging. Hence, the use of conventional propulsion systems is assumed. Initially, at the injection point, the relative velocity in the normal direction must be set to zero, and the radial velocity varies based on the relative distance in the radial direction, which should be equal to the product of the target's mean motion and the relative distance in the radial direction. For example, if the mean motion of the debris orbit defined in Table 2 is 13.7017 rev/day, this value is equivalent to $9.642 \times 10^{-4}$ rad/s when converted to the appropriate units. If the relative distance in the radial direction at the injection point is 250 m, the relative velocity in the radial direction should be adjusted to 0.2491 m/s $\left(= 9.642 \times 10^{-4} \times 250\right)$. Through the injection maneuver, the spacecraft generates the WSE trajectory, and the target is approached along the transverse direction.

The transverse velocity approaching the target is determined by the walking rate, as depicted in Fig. 11. In this study, it has been set at 150 m/rev. Thus, if only 0.5 revolutions are completed, the spacecraft will have traversed approximately 1,100 m, while after 1 revolution, it will have moved a distance equivalent to the magnitude of the rate, which is 150 m. The magnitude of the walking rate can be adjusted to larger or smaller values depending on the transverse distance at the WSE injection point, thereby enabling the adjustment of the timeline for approaching the target. From the point where the proximity operation trajectory is generated, the target is recognized by the spacecraft, and the relative position is estimated through a navigation filter, allowing for the tracking of the target.

When the transverse distance to the target is approximately 500 m, maneuvers are executed to generate a smaller safety ellipse WSE. This process resembles the initial WSE injection, aiming to reduce the size of the safety ellipse by decreasing the magnitude of the radial relative velocity at the injection point. Assuming a size of 60 m for the smaller safety ellipse, the required magnitude of the radial relative velocity is 0.0578 m/s $\left(= 9.642 \times 10^{-4} \times 60\right)$. Figure 12 illustrates the trajectory of this process. Burn #3 signifies the maneuver to reach the injection position for the smaller safety ellipse, while burn #4 represents the maneuver to match the magnitude of the relative velocity for WSE injection.

After the safety ellipse has been reduced in size, the walking rate is set to zero to ensure the maintenance of the relative transverse distance with the target. Figure 13 illustrates the transfer from the safety ellipse to the position for the final approach. In this scenario, a v-bar approach is assumed for the final approach, and burn #5 signifies the maneuver to position the spacecraft 30 m away from the target in the radial direction. It is crucial to nullify the relative velocity with the target before executing the v-bar approach to maintain the relative position continuously. The maneuver for this

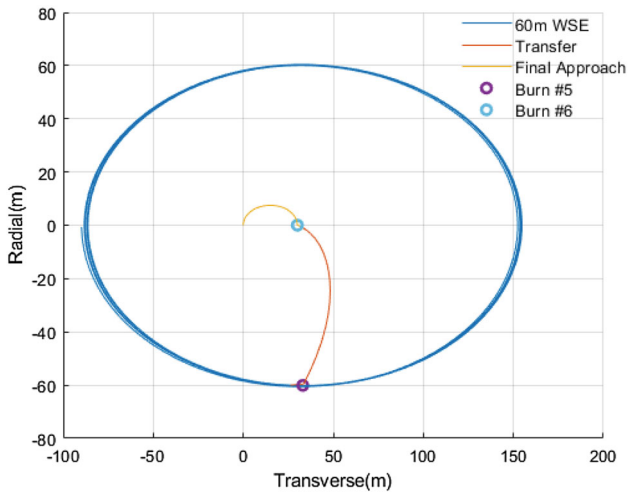


**Fig. 13** Transfer of final approach point and final approach

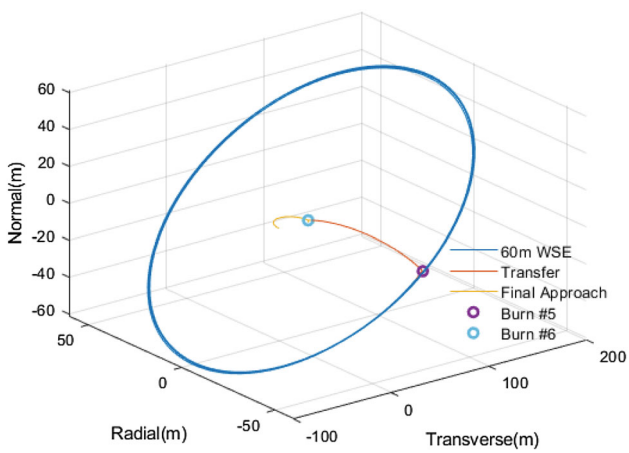


**Fig. 14** 3D trajectory of reduced-size WSE and the final approach

purpose is represented by burn #6. However, due to perturbation forces acting on both satellites, periodic maneuvers are necessary to maintain the relative position. Following the hold state, the spacecraft proceeds with the final approach to capture the target, gradually computing the relative position of the target. Typically, during the approach phase, the approach velocity is controlled to maintain a level relative to the distance, usually around 0.03 m/s if approaching from a distance of 30 m. Figure 14 depicts the 3D trajectory of the final approach process.

The del-$V$ used is calculated using STK's Astrogator module, with impulsive burns assumed to evaluate the mission's feasibility. Here, an impulsive burn refers to the instantaneous application of a del-$V$ value at the thrust activation time. The del-$V$ values used for each stage are summarized in Table 3, with the total del-$V$ required for generating the proximity operations trajectory being 1.8749 m/s. If altitude increase from 600 to 1000 km is performed using Hohmann transfers, the required del-$V$ usage amounts to 207.6879 m/s,

**Table 3** Summary of del-$V$ requirements for proximity operation

| Stage | Del-$V$ (m/s) |
|---|---|
| 250 m WSE transfer | 0.2962 |
| 250 m WSE injection | 0.9163 |
| 250 m WSE correction maneuver | 0.0368 |
| 60 m WSE transfer | 0.2037 |
| 60 m WSE injection | 0.1856 |
| 60 m WSE correction maneuver | 0.0039 |
| Final approach transfer | 0.0945 |
| Hold maneuver | 0.314 |
| Final approach | 0.0075 |
| Total | 1.8749 |

which is significantly higher than the del-$V$ required for proximity operations generation. Spacecraft equipped with sails typically aim to reduce mass to increase characteristic acceleration, making it challenging to carry a significant amount of fuel. Therefore, for missions requiring repetitive orbit transfers like multiple debris removal, it is efficient to leverage sails for ascent and descent as much as possible and reserve fuel solely for proximity operations generation.

### 4.3 Deorbit Phase

After the successful capture of the debris by the spacecraft, the deorbit phase commences. At this stage, significant consideration is given to the mass alteration experienced by the spacecraft post-capture. The addition of debris mass to the spacecraft results in a decrease in the characteristic acceleration of the sail. Assuming a debris mass of 10 kg is captured, the total mass increases to 24 kg, leading to a reduction in the $a_c$ to 0.034 mm/s$^2$. Here, the target orbit is assumed to have an altitude of 550 km, with no specific target values set for eccentricity and inclination. The results are depicted in Fig. 15. Due to the decrease in the $a_c$ value, it is observed that lowering the altitude to 550 km takes approximately 200 days. Subsequently, following the release of the debris at the 550 km altitude, the spacecraft proceeds with orbit transfer toward other targets, aligning with the mission concept.

To provide readers with insights into the trend of altitude reduction using the sail, additional simulations are conducted. Initially, the magnitude of $a_c$ is kept fixed at 0.06 mm/s$^2$. Altitude variations are then examined to compare the time required for deorbiting down to 100 km. Subsequently, with the initial altitude set at 1000 km, changes in the $a_c$ magnitude are investigated to determine the deorbit time. The results for each scenario are depicted in Figs. 16 and 17. It is observed that with a constant $a_c$ value of 0.06 mm/s$^2$, reducing the altitude from 1000 km to 100 km takes approximately

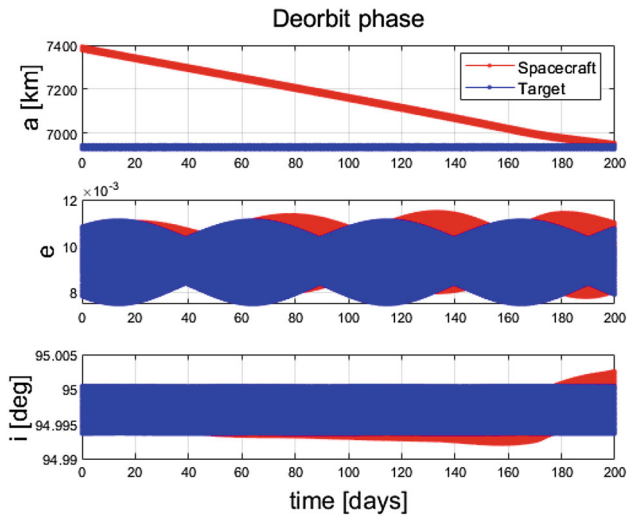


**Fig. 15** Results of deorbit phase simulation outcome

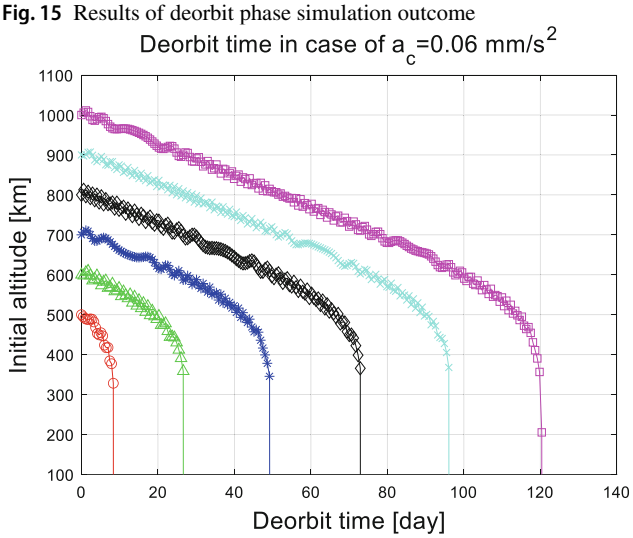


**Fig. 16** Deorbit time with different initial altitude when $a_c = 0.06\,\mathrm{mm/s^2}$

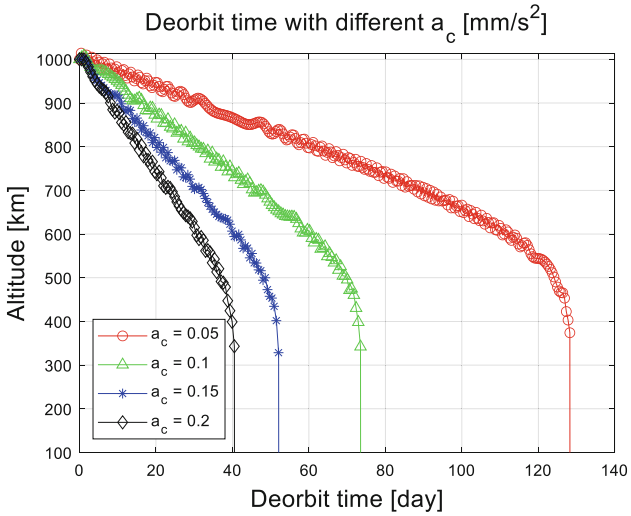


**Fig. 17** Deorbit time variation with $a_c$ at 1000 km altitude

120 days. However, when $a_c$ is increased to 0.1 mm/s$^2$, the time is significantly reduced to around 70 days. Designing the spacecraft with higher $a_c$ values facilitates the release of debris at lower altitudes, expediting the natural decay process. Additionally, the shorter duration required for altitude ascent and descent allows for more frequent, repetitive debris removals. Hence, the advancement in solar sailing technology holds promise for achieving higher values, suggesting ample scope for more diverse and flexible mission designs beyond those presented in this study.

## 5 Conclusion

In this study, a preliminary trajectory design is conducted to perform multiple debris removal missions using solar sail technology. Debris in low Earth orbit is assumed as the target, with consideration given to various perturbations, including atmospheric drag. Additionally, the analysis encompassed not only rendezvous and deorbit processes, but also the consideration of proximity operations for practical capture. Of particular significance is the validation that solar sails, owing to their propellant-free nature, can affect transitions to target orbits for both ascent and descent without fuel consumption. Furthermore, the design process for proximity operations aimed at target capture is elaborated upon, ensuring passive safety through the utilization of the walking safety ellipse. In conclusion, by varying the characteristic acceleration, a critical performance index of solar sails, the feasibility and limitations of the proposed mission concept are assessed. However, considering the potential for future advancements in solar sailing technology to secure higher characteristic acceleration values, it is anticipated that this study will serve as a foundation for presenting a broader array of mission concepts in the future.

## Declarations

**Conflict of interest** On behalf of all authors, the corresponding author states that there is no conflict of interest.

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